

When does cortical coherence carry shared content?

Compatibility and self-reconstruction as testable conditions

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Abstract

Synchrony accounts treat collective phase order as the mechanism that integrates distributed cortical activity. We show formally that phase order alone establishes neither shared content nor a self-model, and state what additionally does. We separate three conditions: coherence (collective phase order), compatibility (agreement of independently decoded local descriptions where regions overlap) and self-reconstruction (accurate recovery of the global state from an internal encoding). Compatible local descriptions determine a unique global state. Coherence bounds disagreement between overlapping descriptions only through shared Lipschitz encoders, and the bound grows as the square root of the population read: at the patch order of a simulated cortical sheet it constrains at most 10 sites, so on a larger population the positive link from coherence to content rests entirely on a bridge assumption. A contracting self-map fixes one state but reconstructs accurately only states within a diameter set by its error and contraction factor; a broad family needs input-conditioned maps

and discriminating codes. Lean 4 verifies these implications, and one model satisfies all eight connecting hypotheses together; this shows consistency, not cortical realization. The conditions yield a disconfirmable comparison: decoders fitted independently on overlapping regions should agree more on reported than on matched unreported stimuli at equal phase coherence. An exploratory EEG analysis is a measurement check only; its pooled estimator places a perfectly synchronized signal at zero coherence. Resource and implementation constraints, and a held-out onset comparison during anaesthetic emergence, are stated briefly. Unity and a minimal self remain interpretations that need an independent experiential endpoint.

1 Introduction

Endogenous extracellular fields can affect spike timing and may coordinate activity across cortical regions (McFadden, 2002, 2020; Pockett, 2012; Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Coordination raises a further question: when do those regions describe one situation, and when does the resulting state represent its own place in it?

Consider reaching for a cup. Vision and movement can describe different features while agreeing on the cup’s location. Matching their timing leaves agreement about that shared quantity untested. An internal description of the perceiver’s relation to the cup asks for still more. We call the three requirements *coherence* (collective phase order), *compatibility* (agreement of local descriptions on overlaps), and *self-representation* (accurate reconstruction from an internal encoding). The mathematical states are spatial measures and fields; identifying them with perceptual content requires independent

15 decoding evidence.

16 Binding-by-synchrony and communication-through-coherence accounts
17 treat phase relations as the medium of integration or of selective routing
18 (Singer & Gray, 1995; Fries, 2015), and field theories place integration in the
19 endogenous electromagnetic field (McFadden, 2020; Pockett, 2012). Both
20 leave open what fact about synchronized activity makes it one content rather
21 than coordinated activity carrying several. The contribution here is to sep-
22 arate the three requirements and prove which follow from which. Phase
23 order constrains shared content only through encoders related on overlaps,
24 with a bound that grows as the square root of the population it is read
25 on. Compatible local descriptions, not synchrony, determine a unique global
26 state. A contracting self-map fixes one state but can reconstruct accurately
27 only a family of bounded diameter. Each result names a measurement that
28 phase data alone cannot supply.

29 The objection that synchrony does not fix content is older than this work.
30 Critiques of the temporal binding hypothesis argued that synchrony, where
31 it occurs, cannot say which features belong together or what they jointly
32 represent: that grouping has to be computed before synchrony could sig-
33 nal it (Shadlen & Movshon, 1999). The formal results here are elementary:
34 uniqueness of gluing, Banach’s fixed-point theorem, the triangle inequality
35 and a pigeonhole count. What formalization adds to the informal argument is
36 quantitative and bookkeeping content. It gives the rate at which a coherence
37 bound on content weakens with population size and the population at which

38 it stops constraining anything, the diameter within which one contracting
39 map can reconstruct, and the number of codes a separated family of states
40 requires. It also gives a complete inventory of the premises a synchrony-to-
41 content argument consumes, each of which names a measurement, together
42 with one model in which all of them hold at once.

43 The familiar noisy mean-field oscillator supplies a stationary coherence
44 threshold (Sakaguchi, 1988; Strogatz & Mirollo, 1991). We add a resource
45 condition for a declared actuator, bounds that connect phase order to con-
46 tent only through specified encoders, and conditions under which compatible
47 descriptions glue and an accurate reconstruction fixes the glued state. A
48 Lean 4 development checks these implications and a joint witness satisfies
49 their eight connecting hypotheses in a specified model. It proves neither that
50 cortex satisfies those hypotheses nor that the glued state is experienced. The
51 latter identification motivates the framework and needs its own endpoint.

52 The empirical question is which measurements could support or discon-
53 firm the proposed connections. A phase resultant can miss a locally or-
54 dered winding field; stationary phase distributions can leave the coupling
55 mechanism unidentified; and coherent adaptation can lose alignment with
56 a designated content template. Independent decoders must identify shared
57 quantities before their agreement can be tested. We therefore distinguish sta-
58 tionary branches from finite observed onset, and compare prespecified field
59 and connectivity onset models on held-out emergence recordings. The com-
60 parison tests a specified spatial prediction, not the presence of experience.

61 Section 10.2 states the comparison that bears on conscious content directly.

62 Sections 2–5 develop the three requirements; Section 6 applies their mea-
 63 surement obligations to a digital candidate. Section 7 states the resource and
 64 composition premises, and Sections 8–9 examine observables and the cortical
 65 test. Table 1 lists the eight connecting hypotheses; Supplemental Material
 gives their derivations and controls.

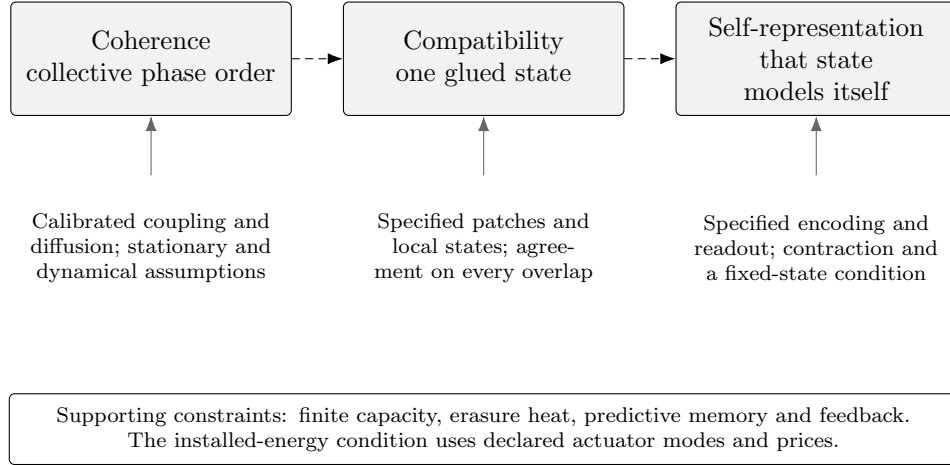


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

66

67 **2 Limits of phase order**

68 Several limits hold without the cortical hypothesis. Equal phases can coexist
69 with incompatible local descriptions; phase order constrains shared content
70 only through encoders related on overlaps. A winding field can have zero
71 global resultant while remaining locally ordered. A constant readout can
72 have a unique fixed point without reconstructing its input. A region that
73 does not distinguish relevant states cannot encode them, and a site cannot
74 report a change outside its causal past by a declared deadline. These lim-
75 its reflect familiar phase oscillator and information constraints (Acebrón et
76 al., 2005; Ott & Antonsen, 2008; Dörfler & Bullo, 2014; Breakspear, 2017).
77 The sections below state each one beside the condition or measurement it
78 motivates.

79 **3 Coherence: physical coupling and its re-** 80 **source condition**

81 **3.1 Dynamics and the stationary threshold**

82 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

83 Here θ_i is the phase of oscillator i , ω_i its natural frequency, $D > 0$ is phase dif-
 84 fusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units,
 85 with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; fre-
 86 quency heterogeneity and diffusion are distinct parameters. The stationary
 87 calculation below uses identical frequencies and this scalar coupling. Ap-
 88 plying it to a spatial kernel requires an aggregation rule and a check of the
 89 resulting approximation.

90 For identical frequencies in their rotating frame, the positive normalized
 91 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

92 where I_n are modified Bessel functions and ψ is the collective phase. The for-
 93 mal derivation uses the absence of stationary current for a periodic gradient
 94 drift to obtain this density, then verifies normalization and self-consistency.
 95 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988; Strogatz &
 96 Mirollo, 1991). Above threshold, exactly one positive solution coexists with
 97 zero and increases strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

98 Figure 2 shows this stationary branch.

99 Linearization about the uniform density gives rate $K/2 - D$ for the first
 100 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists

101 exactly above threshold. About the coherent density, the quadratic form
 102 is strictly dissipative transverse to collective rotation, whose rate is zero.
 103 These linear estimates concern the limiting equation. Relating them to finite
 104 stochastic trajectories requires a dynamical mean-field limit and branch selec-
 105 tion; exchange symmetry alone does not establish the needed independence
 106 (Sznitman, 1991).

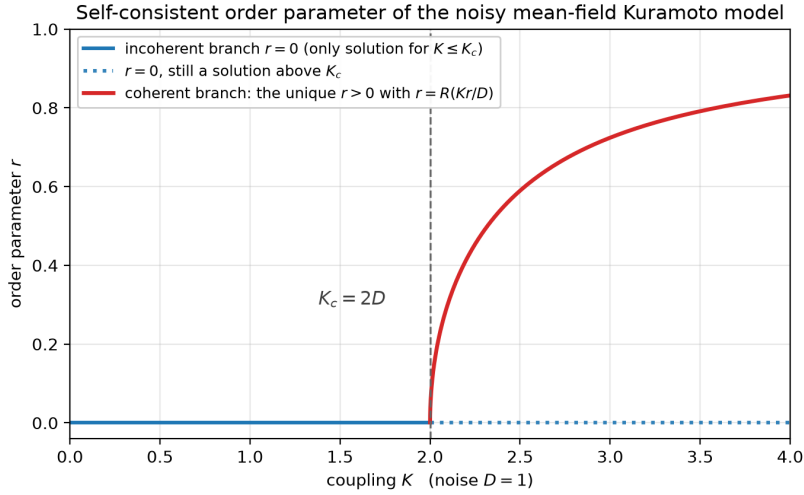


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

107 **3.2 Installed energy bounds mean coupling**

108 A spatial model replaces the finite sum by integration against a kernel. One
 109 declared implementation carries response modes with continuous profiles ϕ_i ,

110 installation prices u_i and nonnegative occupancies $c_i(\Xi)$ in configuration Ξ :

$$K_{\Xi}(x, y) = \sum_i c_i(\Xi) \phi_i(x) \phi_i(y), \quad U(\Xi) = \sum_i u_i c_i(\Xi). \quad (4)$$

111 The same configuration sets the reciprocal kernel and stored energy. An
 112 executed transition changes the kernel through its changed occupancies and
 113 requires work $U(\Xi') - U(\Xi) + Q$, with Q positive into the reservoir. The
 114 installed kernel is averaged over the resulting configuration law $p(\Xi)$. Cell-
 115 pair reconstruction on refining spatial partitions converges to its product-
 116 measure integral under the specified compactness and sampling assumptions;
 117 the refinement index is independent of the number of control updates.

118 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 119 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's
 120 integral gives

$$\begin{aligned} U_{\text{inst}} &= \sum_{\Xi} p(\Xi) U(\Xi), \\ K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_{\Xi} p(\Xi) \sum_i c_i(\Xi) m_i^2 \leq \kappa U_{\text{inst}}. \end{aligned} \quad (5)$$

121 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 122 coherent stationary branch and any growing Fourier mode about the uniform
 123 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 124 is necessary: energy above it need not produce sufficient coupling because
 125 the bound can be loose.

126 The theorem concerns stored energy in the declared actuator modes. The
 127 estimated cortical signalling expenditure and extracellular electrostatic en-
 128 ergy in Supplemental Material measure different quantities; neither identifies
 129 the occupancies, mode prices or their relation to metabolic supply. Conse-
 130 quently the bound supplies a necessary condition for a declared actuator,
 131 but no cortical value of κ or demonstration that cortex meets it. Applying
 132 the scalar threshold to heterogeneous or signed spatial dynamics also needs
 133 a justified reduction.

134 **4 Compatibility: from coordinated activity** 135 **to a global description**

136 **4.1 Why common timing is insufficient**

137 For overlapping patches U and V , compatibility means that restricting either
 138 local description to $U \cap V$ gives the same object. Descriptions can agree on
 139 shared quantities while contributing different information elsewhere. Com-
 140 patible sections of the specified sheaf have a unique global extension. On
 141 a finite cover, local finite spatial measures agreeing as measures on every
 142 overlap also have a unique finite-measure extension whose sheaf image is the
 143 glued section. No mass-one normalization is imposed.

144 In the cup example, visual and reaching descriptions must express a
 145 shared quantity, such as cup location, in common coordinates. Restriction

146 retains that quantity while dropping details particular to each description.
147 Agreement permits surface appearance and hand configuration to contribute
148 to one situation. The mathematical counterexample assigns the same phase
149 to every patch while two local mass profiles disagree on a shared site. Those
150 profiles have no common extension. Because they are chosen independently
151 of phase, synchrony places no constraint on their disagreement.

152 This distinction limits the biological interpretation. Receptive territories
153 and content domains must be related to spatial patches, and the descriptions
154 must support the required extension. Matching overlap marginals of de-
155 coded probability laws need not determine a unique global joint distribution
156 (Abramsky & Brandenburger, 2011).

157 The cover is itself a choice, and the construction constrains it in one way
158 only: synchronized patches agree where they overlap. Where they do not
159 overlap there is nothing to compare, so a cover by pairwise disjoint patches
160 satisfies that obligation for *every* assignment of local descriptions and phases
161 and the gluing theorem still returns a unique global section, which is the
162 tuple it was handed. Membership of the class then reports nothing about the
163 system beyond the patches covering it. What a gluing claim asks of a cover
164 is therefore a non-degeneracy the theorem does not assume: that distinct
165 patches meet, and that what they share is a quantity fixed independently of
166 the agreement it is being used to establish.

167 4.2 When coherence does constrain content

168 A declared relation between phase and content gives a positive result. Let
 169 contents take values in a real normed vector space $\mathcal{C}$. Patch i reads an
 170 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 171 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 172 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 173 sites; their equality on overlaps is a supplied relation between descriptions of
 174 the shared quantity. Write $z_k = e^{i\theta_k}$ and $m = N^{-1} \sum_k z_k$, so $|m| = r$. The
 175 phasors' squared deviations from their mean sum to $\sum_k |z_k - m|^2 = N(1 - r^2)$,
 176 and $|z_i - z_j| \leq |z_i - m| + |z_j - m|$ gives $|z_i - z_j|^2 \leq 2N(1 - r^2)$ for each pair.
 177 Consequently every overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2N(1 - r^2)}. \quad (6)$$

178 Here discrepancy is the norm of the difference, and r and N are read on
 179 whichever population the bound is applied to. That choice is what decides
 180 whether the estimate says anything. A patch carries an order parameter
 181 of its own, and reading the bound there carries that patch's resultant and
 182 that patch's site count, so a large population observed through small patches
 183 is constrained by the small count. Read instead on the whole array, the
 184 factor $\sqrt{N}$ makes the estimate weak, because it draws a pointwise guarantee
 185 from an average over N sites. The order cannot be improved in general:
 186 one oscillator in antiphase among N leaves $N(1 - r^2)$ near 4. At $r = 1$ the

187 observations agree exactly at either scale. A shared scalar encoder is a special
 188 case. Dropping the overlap relation permits disagreement even at perfect
 189 locking. A common nonexpansive linear projection preserves agreement, but
 190 agreement of a projection does not establish agreement of the full vector.

191 What Eq. (6) bounds is a chord, which never exceeds 2, so the estimate
 192 constrains anything only where the right-hand side falls below that. At the
 193 sheet’s own patch order, the bound is informative for at most 10 sites; beyond
 194 that, the coherence-to-content link requires the bridge assumption E78 of
 195 the cortical hypothesis (Table 1 gives the full composition and its edges).
 196 Supplemental Material provides the detailed derivation of this saturation,
 197 and states three conditional derivations that replace the uniform factor $\sqrt{N}$
 198 by a count of disagreeing pairs, a chain across a cover’s shared sites, or a
 199 spectral gap at a frequency-locked configuration.

200 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 201 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 202 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 203 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

204 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 205 duce an initial mismatch. Approximate observations leave a nondecaying
 206 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 207 update to the phase process.

208 4.3 What approximate agreement determines

209 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
210 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
211 Weights that vanish outside their patches and sum to one at each site select
212 a global profile by weighted averaging. The selection lies within ε of each
213 patch there, and any two such selections differ by at most ε , independently
214 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
215 $L\varepsilon$ across selections.

216 At zero tolerance, the selection is the unique exact extension. At every
217 positive tolerance there are families with no exact extension; selection then
218 gives an approximation whose dependence on weights is bounded. Supple-
219 mental Material gives a three-site witness attaining the bound. The mass-
220 profile construction uses the scalar case of the content dynamics and addi-
221 tionally requires nonnegativity of the initial profiles and observations. The
222 general vector result supplies no identification with the spatial-measure sheaf.

223 These constructions concern declared objects: a specified cover, contents
224 valued in a normed space, and encoders related on overlaps. They select
225 neither the neural cover nor the decoder, and identifying their contents with
226 perceptual content, or establishing that the identification bears on experi-
227 ence, is a further modelling task; Section 8 sets out what a measurement of
228 it would have to establish.

229 5 Self-representation: a state that models it- 230 self

231 5.1 The reflexive construction

232 Compatibility gives a global state. Reflexivity relates it to an encoding within
233 that state. Equip global sections with a complete metric. If the additional
234 restriction-resonance condition identifies encoding with restriction to a region
235 $\mathcal{A}$, the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \quad (8)$$

236 The region and readout are fixed across the state space. If F contracts and
237 fixes the section s_* formed from the local descriptions, Banach's theorem
238 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
239 (Table 1 gives the full composition and its edges) supplies the contraction
240 and $F(s_*) = s_*$; restriction resonance is separate. Supercritical coupling
241 makes the stipulated contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ less than one for
242 a relaxation time $\tau > 0$. Constructing a map with that factor remains a
243 modelling obligation. A fixed contraction can reconstruct accurately only a
244 restricted family of states: if its error is at most ε throughout that family,
245 any two members are at most $2\varepsilon/(1 - \Lambda)$ apart.

246 In the cup example, the global state includes the represented scene and the
247 perceiver's relation to it. A local encoding must retain enough information

248 for the readout to recover that modelled state; cup position alone would be
 249 insufficient if hand configuration and viewpoint could not be reconstructed.

250 For the same map with Lipschitz factor $\Lambda < 1$, approximate reconstruc-
 251 tion gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (9)$$

252 At the stipulated rate this radius grows as $2\varepsilon/((K - 2D)\tau)$ near threshold, for
 253 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
 254 weakens.

255 5.2 Reconstruction across a family of states

256 Sensitivity can be required quantitatively. Declare relevant macrostates, an
 257 encoder Φ and readout R , and require $d(s, R(\Phi(s))) \leq \varepsilon$ throughout that
 258 family. If two states have the same code, the triangle inequality gives

$$\Phi(s) = \Phi(t) \implies \max\{d(s, R(\Phi(s))), d(t, R(\Phi(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

259 Thus M relevant states pairwise more than 2ε apart require at least M
 260 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 261 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 262 existing self-prediction map.

263 This is an additional accuracy criterion on a family, with no requirement
 264 for multiple exact fixed points. For the same fixed contracting map used by
 265 E89, however, the diameter bound above is an additional restriction: at zero

error the family contains at most one metric-distinct state. A broad family therefore requires a different, possibly input-conditioned reconstruction map with its own accuracy and stability assumptions. A blind encoder or constant readout fails the stated criterion for separated states despite having a unique fixed point.

Where the encoding is read as the state's own restriction to a region, the criterion becomes a condition on the region. Restriction resonance requires the encoder and that restriction to identify the same pairs of states, so a boundary reconstructing two relevant states more than 2ε apart while its region reads them identically is not resonant. Read forwards, a resonant boundary that reconstructs a pairwise separated family restricts injectively on it. Which states a region reads identically is fixed by where the region sits, before an encoding is chosen, so the resolution the family demands is the region's and no readout supplies it. The implication runs one way: a region that does resolve the family is not thereby shown to carry a resonant encoding, and E89 supplies no such resolution.

Resonance and resolution are jointly satisfiable without a local fold. Take the region to be the whole substrate and use the identity readout: restriction is the identity, so this boundary is resonant and reconstructs every declared family exactly. Its map has Lipschitz factor one and does not satisfy E89's strict contraction premise on a nontrivial metric space. A properness condition on the region is therefore a separate requirement, and the fixed-contraction diameter bound remains in force when E89 is also required. With

289 an L -Lipschitz readout, a family reconstructed within ε whose members lie
 290 pairwise more than $d_{\min}$ apart has δ -separated codes whenever $\delta L \leq d_{\min} - 2\varepsilon$,
 291 so the family is no larger than the δ -packing number of what the region car-
 292 ries. That is the shape a continuous family of states needs, and at the whole
 293 substrate it says nothing, because the region's range is everything. The gain
 294 L is declared and derived here for no physical readout, and at unbounded
 295 gain the bound says nothing.

296 **5.3 Agreement after a change takes communication**

297 Snapshot compatibility does not specify how quickly descriptions respond
 298 when a shared quantity changes. In a synchronous network whose updates
 299 read only the site's own state and declared neighbours' messages, two initial
 300 configurations agreeing on a site's causal past through T rounds produce the
 301 same state there at round T . The deadline consequence is the one Section 2
 302 states, and it follows from the reconstruction bound in Section 5.2: a report
 303 from that site is a readout of a code that the two interventions leave identical.

304 A three-site delay line witnesses failure after one round and exact report-
 305 ing after two; cutting the path prevents delivery at every round. A report
 306 may still happen to be correct about one fixed world. The bound concerns
 307 responsiveness across interventions.

308 Throughout this section *prediction* names a reconstruction map, and the
 309 relevant family of states, the metric, the tolerance, the communication graph
 310 and the deadline are all declared inputs, as are the represented variables and

the fidelity counted as accurate. Reading the map as temporal neural prediction would require dynamics; reading the resulting code count as physical memory capacity would require a hardware model, which a state-dependent readout would have to extend.

6 What the conditions constrain in a digital candidate

The conditions bear on a large language model (LLM) running on graphics processing units (GPUs) only through measurements, and only for a declared task. For a finite spatial coupling operator with eigenvalues $|\lambda_1| \geq \dots \geq |\lambda_N|$, any linear encoder and readout factoring through d real coordinates pays reconstruction error at least $\sum_{i>d} \lambda_i^2$. That rank premise is a property of an explicitly linear bottleneck; query/key dimension does not supply it for softmax attention. On one synthetic two-view task whose target is the glued state and a self-relative coordinate, a rank-two linear decoder has predicted training-error floor 2.000 and held-out error 2.000 on a parity-held-out family. The deadline premise is similarly specific: within one forward pass, the causal mask confines position t to its causal past, so an intervention outside it cannot be reconstructed there by that step. These results rank no architecture family, imply no general inferiority of GPU hardware, and settle no question of machine experience; a claim of unity or self-representation for such a system commits its maker to identifying the shared quantities, the code and its

332 separation of the declared family under intervention. Supplemental Material
 333 gives the bottleneck derivation, the task controls, the information ceiling of
 334 token outputs and the hardware scope.

335 7 Resource premises and composition

336 Two thermodynamic statements enter, with different standing. The first is
 337 proved for the phase model: for a smooth positive periodic density with prob-
 338 ability current J and diffusion D , the current functional $\sigma_J = \int J^2/(D\rho) d\theta$
 339 bounds how fast order can change, $\int_0^\tau \sigma_J dt \geq [\arcsin r(\tau) - \arcsin r(0)]^2/(D\tau)$
 340 while $0 < r < 1$, and stationary order has no positive floor. The second is
 341 conditional: the installed-energy condition of Section 3.2 is necessary for its
 342 declared actuator, and restricts a physical candidate only once mode prices,
 343 coupling reduction and diffusion are measured. Neither is a heat or power
 344 statement until the current is identified with a physical flux, a reservoir at a
 345 stated temperature is specified and local detailed balance holds for the steps
 346 concerned; only then does σ_J convert to entropy production and the finite
 347 register budgets to Landauer heat (Landauer, 1961; Norton, 2005; Shenker,
 348 2000). No composition edge derives coherence, compatibility or reconstruc-
 349 tion from a heat bound. Finite-register capacity obeys $H(p) \leq \log |X|$, and
 350 a separate boundary result concerns a field reaching disconnected vacuum
 351 components (Kibble, 1976); Table 1 states the connecting assumptions. Sup-
 352 plemental Material gives the current-cost derivation, predictive-memory and

353 feedback balances, learning and repeated-operation budgets, and the sensi-
 354 tivity of the metabolic comparison to its assumptions.

355 **7.1 What the composition establishes**

356 The passive and feedback composition theorems each accept eight connecting
 357 implications (Table 1). In the active branch, a named substep at the register’s
 358 thermal scale satisfies

$$k_B T > 0, \quad k_B T [H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (11)$$

359 with B the register’s heat. The upper allocation follows from a ledger whose
 360 operations deliver that total heat to one reservoir and whose other opera-
 361 tions do not increase joint entropy. Those ledger assumptions are physical
 362 inputs. Neither Landauer’s lower bound nor microscopic reversibility sup-
 363 plies them. Finite two-bit witnesses exhibit positive forward and reverse
 364 elementary paths, local detailed balance at one thermal scale, and interac-
 365 tion energies of order $k_B T$; these conditions alone do not quantify proximity
 366 to equilibrium. Cortex has not been shown to satisfy these premises. The
 367 illustrative cortical metabolic expenditure over one region and one phase-
 368 diffusion time is 2.4×10^{14} times kT at body temperature, but identifying it
 369 with the register heat budget B requires a common physical ledger. Read-
 370 ing the register at the scale where its premises are physical, such as a single
 371 channel, requires E45Active to carry that step’s budget to the convergence

372 of a continuum kernel. The active branch is an alternative route to the same
373 node rather than a stronger one, and no result of Section 3.2 passes through
374 it: the installed-energy condition is derived from E56 and E67.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12 <i>capacity and boundary</i>	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23 <i>boundary to register</i>	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active <i>register to dissipation</i>	Bridge assumption	Passive: predictive memory at the register’s thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations’ compressiveness. Both are exhibited at thermal-scale heats on finite state spaces with positive reverse paths, which is not the regime a cortical register occupies.
E45 / E45Active <i>couplings to continuum</i>	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Convergence follows from mesh refinement rather than from the budget, which selects no regime at macroscopic scale. Refinement and physical identifications remain inputs.

Table 1 (continued)

Connection	Status	Required assumption
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours. Where the cover's nerve is connected the same edge carries through the sites its patches share, at a residual linear in that nerve's diameter; where the configuration is frequency-locked it carries through the coupling graph's algebraic connectivity λ_2 instead of through any patch count, which asks the cover for nothing and is declared hardware data in the standing of κ .
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 3.2's occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

375 Both branches also take a thermodynamic cover carrying exact over-
376 lap compatibility and a reflexive boundary specifying encoding and read-
377 out. They conclude that the glued section is the unique fixed point of self-
378 prediction. Compatibility and accuracy at that section enter as premises;
379 uniqueness follows. Identifying encoding with restriction needs the separate
380 resonance condition of Eq. (8).

381 Each of the eight edges is supplied rather than derived from the one above
382 it. E12 is equivalent to its conclusion, so capacity and boundary are inde-
383 pendent roots rather than a step from one to the other; both E45 discharges
384 take their convergence from refinement data and not from the thermody-
385 namic bound; E78 states that coherence relaxes the supplied cover, and the
386 coherence-to-content estimate of Eq. (6) bounds disagreement only for en-
387 coders already shared on overlaps. The inferential content of the composition
388 is its last step, where contraction and a fixed glued section give uniqueness.
389 What composing them establishes is that the premises are consistent, named
390 in one place and machine-checked together, and not that coherence follows
391 from installed energy or self-representation from coherence.

392 Joint witnesses combine a register, boundary, refining mesh and small
393 spatial cover; active witnesses add controlled thermal updates and a priced
394 installation. They establish satisfiability under their stated identifications.
395 Mesh convergence follows from refinement data, and the witnesses establish
396 no common cortical mechanism. Supplemental Material maps the claims to
397 their Lean declarations and records the counterexamples. The development

398 declares no additional axioms beyond its libraries' foundational axioms.

399 8 What the observations establish

400 The composition holds under its premises. Whether observation can estab-
401 lish those premises is a separate question, and the three requirements it
402 connects concern different objects: phase dynamics, local descriptions and
403 internal encodings. Their observables must retain these distinctions. The
404 following controls show which mechanisms remain indistinguishable, which
405 temporal approximations fail, which states one observable cannot separate,
406 and what independent content measurements must establish. Each is a finite
407 numerical regime on a declared model, with full parameters and controls in
408 Supplemental Material, and none of them calibrates a cortical quantity.

409 8.1 Stationary distributions leave the mechanism un- 410 determined

411 Equation (2) gives a concentration-coherence relation with no fitted shape
412 parameter. Estimate r from circular resultant length and a from density
413 shape: inverting R from the measured r would make agreement automatic.
414 Even independently estimated agreement identifies a statistical family shared
415 by different mechanisms.

416 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
417 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$

418 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
 419 oscillators with prescribed drift $-h(t) \sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
 420 and diffusion D_0 have that same density. Their stationary distributions co-
 421 incide window by window despite different causes.

422 Resolved dynamics supply additional information. In the ideal model
 423 with continuous latent paths, quadratic variation accumulates at rate $2D$
 424 and conditional increment means constrain drift; a filtered observation of
 425 the phase needs its own observation model before either is read off. Separat-
 426 ing endogenous coupling from a matched common drive additionally requires
 427 a measured drive or controlled perturbation; identifying electromagnetic cou-
 428 pling requires evidence specific to that interaction. Supplemental Material
 429 gives the equivalence construction and its numerical check.

430 **8.2 Finite ramps change the observable onset**

431 The square-root law in Eq. (3) describes stationary solutions. A changing
 432 coupling requires time for the phase distribution to approach those solutions.
 433 In a deterministic mean-field control that crosses the threshold symmetrically,
 434 none of the scanned rates meets the declared tracking tolerance: slowing
 435 the crossing does not remove the terminal residual under that initialization.
 436 A slow control variable therefore supplies no general guarantee of branch
 437 tracking at onset. Supplemental Material gives the residual, rate range and
 438 admissible rates away from threshold.

439 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas

440 at 8 coupling speeds from $v = 0.1$ to $v = 0.0001$. Escape is the first of three
 441 consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay
 442 escape (Fig. 3); only 5/32 replicas escape within the fastest window. The
 443 6 uncensored legs supply the delay scaling below; the decade-spaced ones
 444 among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted
 445 onset exponents differ from those of the exact stationary branch evaluated
 446 on the same grids. At the fastest speed the ensemble mean never reaches the
 447 fit window, so its fitted slope does not estimate stationary onset.

448 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
 449 makes the apparent exponent depend on ramp speed and fitting window.
 450 The specified deviation-based critical ramp speed is not bracketed by this
 451 sweep. With fluctuating coupling, the relevant stationary comparison also
 452 depends on rate: the mean coupling approximates order in the tested fast
 453 regime, while averaging stationary order over coupling works for the tested
 454 slow drives confined to one side of threshold. Slow crossing drives can satisfy
 455 neither approximation.

456 **The prediction.** Over the 6 uncensored legs a finite-range fit gives $\Delta K \propto$
 457 $v^{0.444}$. The interval from fitting those means is $[0.394, 0.494]$; resampling
 458 replicas independently within each speed gives a 95% interval $[0.398, 0.496]$,
 459 with fraction 0.019 of 10000 bootstrap fits at or above $1/2$. This fraction
 460 describes the resampling distribution under the fitted protocol; it is not a
 461 test of the ideal exponent. A deterministic mean-field run seeded at the

fluctuation floor gives exponent 0.425 with the same estimator and escape
 criterion; tightening its criterion to $r \geq 0.05$ gives 0.490. These controls
 show that the criterion affects the fitted exponent, but do not prove that it
 accounts for the entire finite-population shortfall. The larger-population low-
 threshold fit has substantial log residuals; matched-speed fits do not converge
 toward one-half across the tested sizes and criteria. Some trajectories also
 cross the absolute escape level before K_c , so the first postcritical crossing can
 yield a zero or grid-limited delay. This finite sweep establishes no common
 asymptotic exponent.

The ideal delayed-onset calculation motivates a square-root scaling hy-
 pothesis; its exponent is generic to suitable bifurcations and does not iden-
 tify the coupling medium. A within-subject emergence design can vary the
 propofol taper and test how an operational onset changes with that manipu-
 lation, but interpreting the result as a coupling-rate law requires a measured
 or calibrated relation between taper and $\dot{K}$. The escape criterion must be
 set relative to each recording's pre-threshold fluctuation scale. Section 8.1
 describes how D could be estimated from resolved phase traces.

Additional controls address integration bias, population size and spatial
 or signed coupling; the refined signed-network crossing bracket $K_{\text{eff}}/D \in$
 $[1.958, 2.009]$ contains 2.

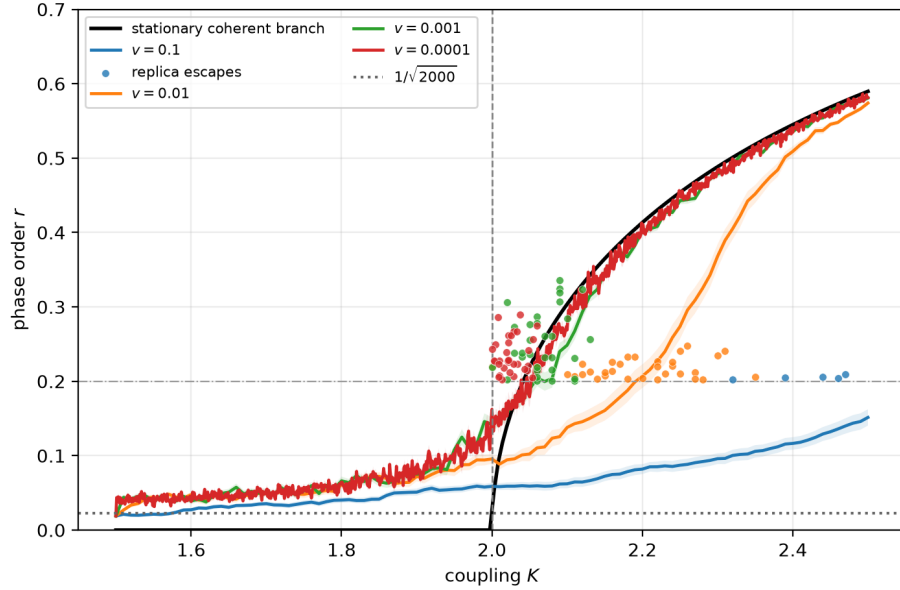


Figure 3: Finite-rate departures from the stationary branch. Curves show ensemble-mean order for four ramp speeds v , with one-standard-error bands. Coloured points mark each replica’s first sustained escape, plotted at its recorded order; censored replicas have no point. The dashed vertical line is the static threshold, the dash-dotted horizontal line is the escape criterion, and the dotted horizontal line is the finite-size reference. The replica-bootstrap interval for the delay exponent is reported in the text.

482 8.3 A winding state leaves the order parameter am- 483 biguous

484 The global order parameter r discards spatial arrangement. On a periodic
485 sheet, a phase field can wind once around the sheet and have $r = 0$ while
486 neighbouring sites remain locked. At identical frequencies, every such wind-
487 ing is stationary for an isotropic symmetric kernel: contributions at opposite
488 separations cancel. It violates the quarter-turn initial-spread premise of the

489 convergence result, which is why that result can still select the uniform state.

490 In the spatial control, the winding persists below the interpolated de-
 491 cay length 0.1871 mm and unwinds above it (Fig. 4). At 0.1 mm, uniform
 492 and winding starts under the same coupling, diffusion and kernel end at
 493 $r = 0.9657$ and $r = 0.0083$; patch-local order is 0.9015. Under quenched fre-
 494 quency spread 1.5 rad s^{-1} , the measured retention boundary is 0.1421 mm,
 495 and the corresponding order comparison is 0.9418 against 0.0348 over 3 frozen
 496 realizations. These are finite controls, not a cortical wave identification or a
 497 derived transition.

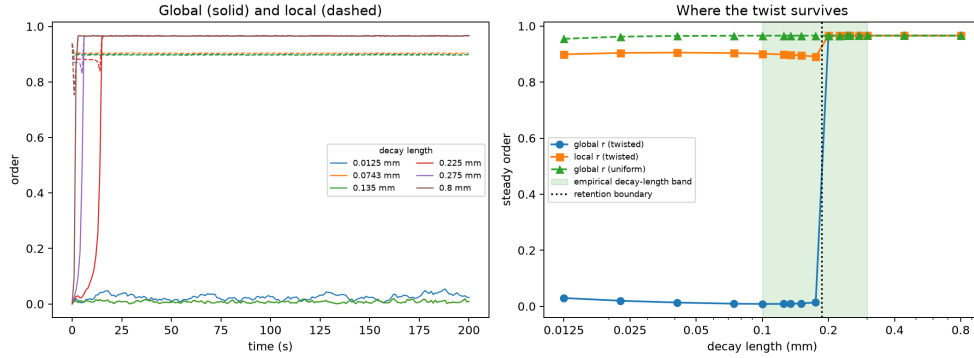


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

498 Linearization about a winding reweights each separation’s coupling by

499 the cosine of its phase advance. The resulting symmetric kernel has explicit
 500 mode rates: nonnegative reweighted coupling gives no growing mode, while
 501 a positive rate destabilizes that winding. The criterion concerns linear rates,
 502 not asymptotic stability, and the measured retention boundary need not
 503 equal its limit. An amplitude extension adds an existence condition: $\Gamma + \lambda >$
 504 0 , where Γ is linear gain and $\lambda = \sum_d f_d(\cos \Delta\theta_d - 1)$ comes from the same
 505 kernel. For the nonnegative kernels used here, a broad kernel can remove a
 506 winding from that model even though the phase-only equations retain it as
 507 a stationary state.

508 Neither model turns these states into measured cortical spirals. The pe-
 509 riodic sheet cannot carry a lone phase defect, the relaxational dynamics has
 510 no propagation speed, and the amplitude variable has no calibrated relation
 511 to a recorded envelope. Supplemental Material gives the spectral, amplitude
 512 and defect results. Locally, Eq. (6) can still bound content disagreement on a
 513 winding patch when shared encoders are supplied, although its global $r = 0$
 514 form is uninformative. A nonzero ring winding number q also obstructs a
 515 global real-valued phase lift; it has no identified content meaning here. Thus
 516 an onset criterion based only on r can register a change of basin as well
 517 as a change of coupling. The protocol of Section 9.3 needs a spatial phase
 518 observable.

519 8.4 Coherent adaptation can lose environmental align- 520 ment

521 An adaptive coupling rule might supply representational structure, but its
522 objective must reward the relevant agreement. For symmetric coupling, pair-
523 wise sine terms cancel in the population mean drift. Minimizing squared drift
524 at that fixed mean therefore rewards equalizing drift across nodes. In the
525 frequency-cluster control, this favors coupling between unlike-frequency clus-
526 ters, whereas the diagnostic environmental template rewards coupling within
527 clusters.

528 The formalization proves descent of the squared-drift objective for a spec-
529 ified gradient flow at fixed phases. Its functional differs from thermodynamic
530 entropy production: at a coherent stationary density the probability current
531 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
532 rial gives the operator, its gradient and the conditions of the descent result.

533 The finite control co-evolves noisy phases and symmetric, nonnegative
534 couplings at fixed total resource. Its template is a diagnostic covariance
535 of latent signals, never a training input. Across three illustrative seeds,
536 second-half order remains between 0.9167 and 0.9222, while the within-
537 cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–
538 0.6135 (Fig. 5). Random-update, permuted-gradient and frozen-kernel con-
539 trols are reported in Supplemental Material.

540 The immediate post-update objective reduction disappears when aver-

aged over complete update intervals. A prespecified grid of 6 rate–cadence combinations contains no case meeting the 5% sustained-reduction criterion while retaining coherence and losing alignment. The finding is coherent adaptation away from the specified structure, with no demonstrated sustained cost reduction.

Kernel alignment and overlap compatibility are different observables. This control addresses acquisition of its designated template under its particular objective. Learning models can explicitly include data in an objective (Corberi et al., 2026); an account of agreement about the cup must specify the content its learning rule is meant to preserve.

8.5 Content agreement needs independent decoders

A neural test needs independently identified domains, local descriptions and restriction maps. Two decoders should estimate a shared content on overlapping domains before their agreement is examined. Discrepancy requires a null under decoding noise and controls using deliberately compatible and incompatible examples. An implementation on synthetic data separates these cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by trial; the compatible construction scores 0.476 on an interval containing one half. Shared-prior shrinkage can nevertheless manufacture agreement, and regional bias can manufacture disagreement. Per-patch decoding diagnostics detect these failures; the compatibility statistic alone does not.

Phase agreement cannot replace that measurement. On constructed con-

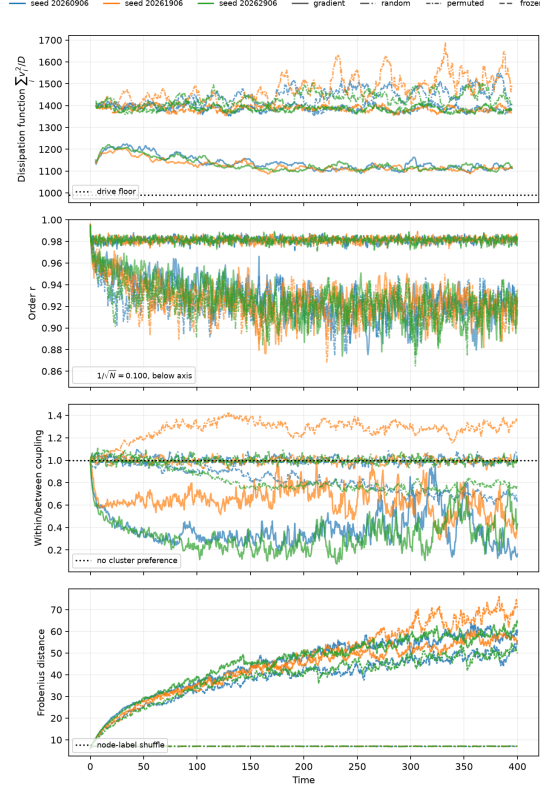


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

563 tent, phase carries a share 1.000 of the content entropy when content is built
564 from it and 0.000 when it is not. A phase-derived statistic reports com-
565 patibility on phase-locked patches whose local descriptions disagree. The
566 implemented content statistic detects their incompatibility. These controls
567 validate a measurement on known constructions; a neural content model and
568 decoder remain to be established.

569 Choosing the cover also chooses a candidate subject: trivial descriptions
570 can agree while carrying little content, and a partition can conceal disagree-
571 ment. The useful choice must therefore be constrained by what neural ac-
572 tivity encodes. The exact composition assumes compatibility of its supplied
573 cover. The content mechanisms above identify additional hypotheses under
574 which agreement can arise; they leave the empirical choice of content and
575 the connection to that cover explicit.

576 **8.6 Estimator calibration on exploratory EEG**

577 The concentration range and the dependence structure decide whether the
578 concentration-coherence relation can be tested at all. Under the specified
579 log-density estimator and bias-floor criterion, discrimination at $a = 0.542$
580 requires 3000 independent sites in the scanned designs, rising to 31000 when
581 sites share a mean in groups of 10. These counts describe that estimator and
582 dependence model at a declared design concentration; the intended recording
583 requires its own calibration. Supplemental Material reports the bin-count
584 controls and the full dependence-aware design.

585 The estimators were also exercised on public propofol-sedation EEG (Ba-
 586 jwa et al., 2025): 61 adjacent bipolar pairs formed from 62 scalp channels,
 587 band-passed, Hilbert-transformed and pooled over pairs and 100-ms win-
 588 dows. Across 8 subjects the pooled concentration spans 0.087–0.142, and
 589 the residual from the Bessel relation is small. That agreement is uninforma-
 590 tive. Pooling over a window does not remove the common rotation of the
 591 phases, so a perfectly phase-locked 10-Hz signal passed through the same
 592 pipeline returns $r = 0.0001$ and $a = 0.0004$ with zero residual, although its
 593 instantaneous spatial resultant is one. The bipolar montage also removes
 594 spatial coherence: a field whose raw channels have instantaneous resultant
 595 0.89 gives 0.06 across bipolar pairs and 0.007 pooled. Any nearly uniform
 596 pooled distribution satisfies $r \approx a/2$, and over the observed range the Bessel
 597 ratio departs from $a/2$ by at most 0.0002. The exercise therefore checks
 598 the estimator’s behaviour and establishes no spatial concentration, coher-
 599 ence level or test of the stationary density. Sedation recordings also leave the
 600 awakening hypothesis untested.

601 An informative dataset therefore needs an estimand that removes the
 602 common rotation, a montage whose effect on spatial phase is modelled, suf-
 603 ficient concentration range or precision, calibrated estimators and an ob-
 604 servation model for dependence and source mixing. Intracranial recordings
 605 (ECoG, sEEG or LFP) or narrowband burst analysis may reach a more
 606 informative concentration range, but that range must be measured. The
 607 design counts above are calculations for the specified estimator, not sam-

608 ple requirements transferable to another recording. Public mouse calcium
609 imaging provides thousands of spatially located neurons across waking, sleep
610 and anesthesia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the
611 phase-estimation band used here. Site count, state coverage and bandwidth
612 are separate requirements.

613 **9 The cortical hypothesis and an awakening** 614 **test**

615 The cortical proposal identifies a physical source of coupling and asks whether
616 its changes help explain recovery. The tests in Section 8 determine what
617 observations must distinguish before a phase trace can support this identifi-
618 cation.

619 **9.1 Physical identification and calibration**

620 The cortical hypothesis identifies part of effective coupling with endogenous
621 extracellular electromagnetic interactions. Fields can modulate spike timing
622 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
623 propose coordination of distributed neural information (Pinotsis & Miller,
624 2022; McFadden, 2020). Phase organization under propofol motivates study-
625 ing this regime (Bastos et al., 2021), but does not measure coupling.

626 The formal field predicate requires a jointly continuous kernel, a normal-

627 ized substrate, coupling equal to the kernel average, and no single site pro-
628 viding more than half that strength. Toy constructions satisfy it. A cortical
629 realization must specify the scale at which the kernel and its reduction ap-
630 ply: calcium imaging reports different state-dependent modularity patterns
631 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

632 Cortical columns are the obvious candidate unit, and they answer one
633 of the two demands the formalism makes. As the oscillator index they are
634 a serious proposal: they are delineated by response properties and anatomy
635 rather than by the agreement they would be used to establish (Mountcastle,
636 1997), and a phase can be estimated from a column-sized territory at the
637 electrode spacings used for cortical wave measurements (Townsend et al.,
638 2015). As the content cover they are a poor one, for three separate reasons.
639 A parcellation into non-overlapping columns is a disjoint cover, which satisfies
640 the compatibility obligation without constraining anything (Section 4). The
641 cover condition additionally requires the patches to exhaust the substrate,
642 while much of association cortex has no accepted columnar parcellation. And
643 the column’s standing as a canonical cortical unit is contested (Horton &
644 Adams, 2005; Rakic, 2008). Offering columns for both roles at once therefore
645 supplies the oscillator index; the content cover remains an open empirical
646 question, which the bridge assumption E78 records.

647 The spatial reduction is also testable. A row-sum aggregation at zero
648 diagonal tracks the uniform-coupling control over the tested spatial decay
649 range, while frequency heterogeneity displaces the operational threshold. The

650 fit window biases both estimates relative to $2D$; agreement with the control
651 is therefore the supported conclusion. Supplemental Material reports the
652 reduction, its numerical error and the distinct assumptions needed for the
653 kernel-average identification in Eq. (5).

654 The field commitment requires physical evidence beyond a choice of math-
655 ematical representation, and the available support comparisons do not supply
656 it. A finite matrix spread over positive-mass cells preserves its total cou-
657 pling and phase correlation exactly, while point-supported vanishing under
658 an atomless measure and a disadvantage from excluding a better-correlated
659 pair concern specified support constraints; Section 6 reads the same compar-
660 isons against a digital candidate. Neither supplies energy prices.

661 A spike-time shift alone supplies no coupling rate. The dimensionless
662 product of population size N , timing sensitivity $\mathcal{T}$, field amplitude E and
663 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (12)$$

664 This rate describes how induced timing shifts accumulate into drift. Nei-
665 ther it nor the installation factor κ follows from a heat bound. Testing the
666 physical threshold requires independent estimates of the interaction kernel,
667 its aggregation and diffusion in the same cortical regime.

668 9.2 Extracellular geometry and recovery

669 The proposed control variable is extracellular geometry. Extracellular vol-
670 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
671 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
672 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
673 at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
674 an effective-medium description predicts that a larger extracellular fraction
675 raises conductivity and reduces field amplitude.

676 The coupling-gated hypothesis consequently predicts that extracellular
677 contraction on awakening raises E and, holding the other factors in Eq. (12)
678 fixed, raises effective coupling. This argument supplies a direction; the
679 geometry–coupling relation, its magnitude and its role in human recovery
680 require measurement.

681 For a smooth crossing at nonzero speed, evaluating the stationary branch
682 along the changing control gives a deterministic tracking approximation that
683 fails at onset in the tested crossing control:

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (13)$$

684 This is a deterministic tracking approximation from Eq. (3) at fixed D ; the
685 crossing controls in Section 8.2 show that it fails as an observed-onset ap-
686 proximation under the tested initialization. A separation of geometry and
687 phase time scales can support tracking away from threshold. The awakening

688 hypothesis needs a calibrated dynamical prediction for an operational onset
689 above the finite-size fluctuation floor. Crossing K_c alone does not supply a
690 recovery latency.

691 **9.3 A protocol for the awakening experiment**

692 Neural inertia describes a behavioural barrier and hysteresis between anaes-
693 thetic induction and emergence (Friedman et al., 2010); recovery can pass
694 through metastable activity states (Hudson et al., 2014; Proekt & Hudson,
695 2018). The $v^{1/2}$ exponent of an ideal delayed bifurcation is generic (Baer
696 et al., 1989; Berglund & Gentz, 2002) and cannot alone discriminate a field
697 mechanism from a synaptic one. A possible discriminator is spatial onset
698 relative to independently estimated local field coupling and network connec-
699 tivity. The present mean-field model has no spatially varying κ and derives
700 no nucleation order, so a field-based onset map needs a separate model spec-
701 ified before the observations are inspected.

702 For that comparison, partition the recording into prespecified regions i
703 and measure pre-onset order $r_i(t_0)$ and diffusion D_i . Define a field score
704 $h_i = N_i \mathcal{T}_i E_i f_i$ from independent pre-onset field and timing measurements,
705 and a connectivity score c_i from a separately estimated structural network.
706 For the same measured emergence drive $g(t)$, compare $K_i^F(t) = a_F g(t) h_i$ with
707 $K_i^C(t) = a_C g(t) c_i$. Each model has one fitted scale. The local first-harmonic

708 approximation predicts

$$\widehat{r}_i^M(t) = r_i(t_0) \exp\left(\int_{t_0}^t [K_i^M(s)/2 - D_i] ds\right), \quad M \in \{F, C\}. \quad (14)$$

709 Predicted onset is the first crossing of a region-specific threshold fixed from
710 pre-onset fluctuations. Fit the two scales on training sessions, then com-
711 pare held-out onset times and site order under the same observation model.
712 Larger field scores predict earlier onset only when baseline order and diffu-
713 sion do not reverse the ranking. This declared approximation omits interre-
714 gional coupling and saturation; a successful ranking would support its field
715 covariate, not prove ephaptic causation. Supplemental Material specifies the
716 controls and failure conditions.

717 A within-subject protocol can manipulate emergence rate, with spatially
718 resolved recordings to compare the proposed field and connectivity models:

- 719 1. **Manipulate emergence rate.** Vary the propofol infusion taper
720 across sessions within the same subject, recording drug concentrations
721 or a validated pharmacokinetic proxy to quantify the manipulation.
- 722 2. **Measure behavioural recovery.** Record time-to-response to a stan-
723 dard stimulus as the independent behavioural endpoint, rather than
724 estimating ΔK directly.
- 725 3. **Resolve spatial onset.** Identify anatomical locations where coher-
726 ence first crosses the baseline-scaled threshold and score the two maps

727 of Eq. (14) on held-out sessions. The two covariate maps are fixed
728 before the outcome sessions are inspected.

729 **4. Match physical controls.** Confirm that the observed coherence-to-
730 amplitude relation holds under the specific local diffusion D and ob-
731 servation noise. The model’s quadratic-variation rate $2D$ applies to
732 the latent phase; a band-limited analytic phase is smooth below its
733 correlation time, and on a constructed field its squared-increment rate
734 changes 37-fold between 0.2-ms and 20-ms steps. Estimating D there-
735 fore requires an explicit observation model for filtering and sampling.

736 Predeclare the recording window, regional fluctuation thresholds, and a
737 shared timing-error model. Fit the two scales and that observation model
738 on training sessions only; on held-out sessions, score the observed site-onset
739 times and sites that never cross by the same regional predictive rule. Report
740 site order as a secondary outcome and behavioural time-to-response sepa-
741 rately. Sessions without a crossing are censored observations, not zero delays
742 or exclusions. Compare the two scores only where independently estimated
743 field and connectivity maps predict distinguishable onset patterns at the
744 recording resolution. A held-out field-model advantage would support that
745 specified spatial predictor; it would not identify an ephaptic cause, shared
746 content or experience. Supplemental Material gives the controls and pilot
747 quantities needed to judge whether the comparison is informative.

748 **10 Discussion**

749 **10.1 Unity and a minimal self as interpretive motiva-** 750 **tion**

751 We propose the state glued from compatible local descriptions as a formal
752 correlate of conscious unity, and its accurate internal reconstruction as a cor-
753 relate of a minimal self in the sense of a pre-reflective point of view on the
754 scene (Gallagher, 2000). In the cup example, this would mean one description
755 that preserves the cup’s relation to the perceiver as well as the scene. Com-
756 patibility permits distinct contents and uncertainty; the shared quantities
757 must be chosen from independent evidence about what each region encodes.
758 The reconstruction need not be a verbal or autobiographical account. Its
759 proposed relation to experienced perspective remains an interpretation. Nei-
760 ther identification is among the eight hypotheses in Table 1, and satisfying
761 them would not measure experience.

762 **10.2 A comparison that could disconfirm the link to** 763 **content**

764 The framework predicts that conscious content tracks compatibility, not co-
765 herence. Present matched stimuli that are reported on some trials and not
766 on others, for example at masking or rivalry threshold, and fit decoders for a
767 shared stimulus quantity independently on two overlapping cortical regions.

768 Compare decoded agreement on the overlap between reported and unreported
769 trials after matching their phase coherence. The proposed link predicts higher
770 agreement on reported trials at equal coherence. Equal agreement, or agree-
771 ment that follows coherence and not report, counts against it. A self-model
772 comparison has the same form: reconstruction of perceiver-relative variables,
773 such as hand position relative to the target, from a declared local encoding
774 should be more accurate on reported trials. These are correlational tests of
775 the proposed correlates; report is itself an access measure, and a no-report
776 design would be needed to separate content from access.

777 **10.3 What a test of the full proposal must establish**

778 A test needs separate estimates of effective coupling and diffusion, shared
779 content on overlapping regions, and reconstruction from a specified encoding
780 at a declared fidelity. The cover, relevant state family and decoder must
781 be fixed without selecting them for agreement. An independent experiential
782 endpoint is needed to evaluate the interpretation; absence of a report alone
783 does not define absence of experience. A unified episode with incompati-
784 ble decoded descriptions would challenge the unity identification. A unified
785 episode lacking the proposed self-encoding would challenge the minimal-self
786 claim. Conversely, satisfying the mathematical conditions without such an
787 episode would challenge their sufficiency as correlates.

788 For the physical route, cortical actuator modes and prices, the geometry-
789 coupling relation and a dynamical onset model remain to be measured. The

790 field/connectivity comparison in Section 9.3 evaluates one spatial onset pre-
791 diction. Controlled changes in coordination and independently decoded con-
792 tent agreement would address the representational connection. The synthetic
793 decoder exercise calibrates part of these measurements, and the EEG exercise
794 identifies what a pooled bipolar estimand cannot measure; neither identifies
795 a neural content code or tests experience.

796 10.4 Explanatory scope

797 Integrated-information, global-workspace and predictive-processing accounts
798 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
799 1997; Dehaene et al., 2014; Friston, 2010). This account specifies conditional
800 relations among coupling, local content and reconstruction. Its formal field
801 condition requires a continuous kernel, a normalized substrate, a specified
802 average coupling and noise, and a site-contribution bound. Those conditions
803 do not identify an electromagnetic substrate; the extracellular interpretation
804 needs the separate calibration of Section 9. The joint witness shows that
805 the formal hypotheses can hold together in a declared model. A cortical
806 realization and an experiential interpretation require additional evidence.

807 Three limitations bound every claim above. Synchrony is not itself an
808 experiential measure, and nothing here establishes a consciousness criterion:
809 the conditions are proposed correlates to be tested against report or another
810 independent endpoint. The present EEG analysis checks a measurement
811 pipeline and does not validate any experiential claim. The formal results are

812 conditional on declared encoders, covers and maps whose cortical counter-
813 parts are unmeasured.

814 The remaining explanatory question is whether these relations occur in
815 neural activity and matter for experience. The framework identifies candidate
816 correlates and tests of their physical and representational premises. It does
817 not explain why satisfying them would be accompanied by experience.

818 11 Conclusions

819 Phase coherence, compatibility and self-reconstruction are separable condi-
820 tions. Coherence constrains shared content only through encoders shared on
821 overlaps and with a bound that grows as the square root of the population
822 read. That bound is vacuous beyond 10 sites at the simulated sheet's patch
823 order, so for populations of cortical size the positive link from coherence
824 to content is the bridge assumption E78 and not a derived result. Com-
825 patible local descriptions, not synchrony, determine one global state; and a
826 contracting self-map fixes that state while reconstructing only a family of
827 bounded diameter. Each result replaces an assumption about synchrony by
828 a measurement: independently decoded agreement on overlaps, a declared
829 encoding and its fidelity, and calibrated coupling and diffusion. The compar-
830 ison of Section 10.2 states how such measurements could count against the
831 proposed link to conscious content.

Notation

Table 2: Symbols used throughout, with the section that defines each. Where one letter carries more than one meaning, the meanings are separated by the section they are read in.

Symbol	Defined	Meaning
θ_i, ω_i	Eq. (1)	Phase and natural frequency of oscillator i
N	Eq. (1)	Number of oscillators in the population the quantity is read on
D	Eq. (1), Supp.	Phase diffusion in inverse time; in the Supplemental Material's convergence proof, the phase spread $D \leq \pi/2$
K_{ij}, K	Eq. (1)	Pairwise and scalar mean-field coupling, $K_{ij} = K/N$
K_c	Eq. (2)	Synchronization threshold, $K_c = 2D$
r, r_N	Eq. (1)	Resultant length (order parameter), on the population it is read on
a, ψ	Eq. (2)	Concentration $a = Kr/D$ and collective phase of the stationary density
q	Section 8.3	Winding number of a phase field read around a ring
μ	Eq. (4)	Substrate measure, normalized to one
Ξ, ϕ_i, u_i, c_i	Eq. (4)	Actuator configuration; mode profile, installation price and occupancy of mode i
U_{inst}, κ	Eq. (5)	Installed energy and the mode-data conversion factor bounding $K_{\text{mf}} \leq \kappa U_{\text{inst}}$
L	Section 4.2	Lipschitz constant of an encoder or readout
ε, δ	Section 4.2	Content and observation discrepancy tolerances
M, b	Eq. (10)	Size of a declared family of states and code length in bits
Λ, τ	Eq. (8)	Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time
T	§7, §5	Temperature in the thermal scale $k_B T$; number of communication rounds to a deadline
ν	Supplemental Material	Fan-in of a communication graph: the most sites any one site hears from

Table 2 (continued)

Symbol	Defined	Meaning
$k_B T, B$	Eq. (11)	Thermal scale, read as one symbol, and heat budget of a register
$\gamma, \mathcal{T}, f$	Eq. (12)	Field-to-coupling conversion rate, timing sensitivity and frequency
α	Section 9	Extracellular volume fraction
λ	§8.3, §9	Eigenvalue the coupling kernel supplies at a winding; extracellular tortuosity
Ω, λ_2	Section 4.2	Common rotation rate of a frequency-locked configuration, and the algebraic connectivity of the coupling graph
β, k	Section 4.2	Chord diameter of a patch and the diameter of a cover’s nerve, in hops
$\mathcal{C}, e_{i,x}$	Section 4.2	Content space and the phase encoder patch i reads at site x
Φ, R	Eq. (10)	Encoder and readout of the reconstruction criterion
$\mathcal{A}, F$	Eq. (8)	Encoding region and the reconstruction map $F(s) = \text{readout}(s _{\mathcal{A}})$
z_i, Γ, A_{ij}	Section 8.3	Complex amplitude, linear gain and coupling matrix of the amplitude layer
$\Delta\theta_d$	Section 8.3	Phase a winding advances across separation d
E	Eq. (12)	Field amplitude, in the cortical identification
U	§3.2	Stored energy $U(\Xi)$ of a configuration
s, s_*	§4–5	A global section and the one the reconstruction fixes

833 Data and code availability

834 The Lean development, simulation scripts, tests and saved numerical sum-
835 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
836 `cortical_coherence`. The formalization uses Lean 4.34.0-rc2 and Math-

lib commit 5ce203ecd53f8c3e47ee5530d7c86b94a55f0183. Publication
simulation values are generated from saved summaries without rerunning
production sweeps. The exploratory EEG analysis uses OpenNeuro dataset
ds005620 (Bajwa et al., 2025) (subjects 1010, 1016, 1022, 1033, 1045,
1054, 1060, 1067, resting awake and sedation runs 1–2); preprocessing
and selection are described in Supplemental Material and implemented in
simulations/empirical_collapse.py. The repository contains the repro-
ducible analysis and does not redistribute the raw recordings.

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Use of AI tools

AI assistance to this work was substantial and is not separable by task.
Several coding assistants and chat interfaces were used in an interleaved way
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Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra and Sol; GPT-
5.6 Terra, Luna and Sol), Claude Code (Claude Opus 5 and Opus 5.5),
Gemini Pro through its web interface, and Anthropic’s Fable and DeepSeek
v4 Flash through OpenRouter, accessed with the Hermes CLI. Between them

856 they contributed to the Lean development, the simulation and analysis code,
857 and the drafting and revision of both documents. Because the tools were
858 used in combination and in alternation, no section, proof, figure or result is
859 attributable to any one of them, and none is claimed to be. The author is
860 responsible for all content, including everything a tool produced. Machine
861 checking establishes the stated Lean results under their hypotheses; it does
862 not validate biological interpretations or replace scientific review of generated
863 code and prose.

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